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┌────────────────── MODULE instructions_ooo ───────────────────┐
EXTENDS Integers, Sequences
CONSTANTS varIF, N_FU, Lat, missLat, superscal, modeLen, Program
└──────────────────┘

ProgLen  $\triangleq$  IF modeLen = - 1 THEN Len(Program) ELSE modeLen

empty  $\triangleq$  [pc  $\mapsto$  0, type  $\mapsto$  “empty”]

next_instr(p, i)  $\triangleq$  IF Len(p) > i - 1 THEN p[i] ELSE empty

RECURSIVE rest_instr(-, -)
rest_instr(p, n)  $\triangleq$  LET rest(l)  $\triangleq$  IF Len(l) > 0 THEN Tail(l) ELSE  $\langle \rangle$  IN
    IF n = 1
    THEN rest(p)
    ELSE rest_instr(rest(p), n - 1)

exec_instr(p, pc)  $\triangleq$  IF pc.PC  $\neq$  empty THEN Append(p, pc) ELSE p
└──────────────────┘

```